

WEIGHTED THREE-TERM SHIFTS ON FINITE HEISENBERG QUOTIENTS

ANONYMOUS

ABSTRACT. We compute the congruence-fixed spaces of a weighted three-term family of linear shifts over the discrete Heisenberg group. Let ℓ be an odd prime, let $p \neq \ell$ be prime, and reduce the group modulo ℓ . For nonzero coefficients $\alpha, \beta, \gamma \in \mathbb{F}_p$, the fixed-space dimension is the sum of a cyclotomic torsion-intersection degree and a nonlinear term. The nonlinear term is either zero or the full jump $\ell(\ell - 1)$, and it occurs exactly on the Fermat locus $\alpha^\ell + \beta^\ell + \gamma^\ell = 0$. The proof decomposes the finite Heisenberg regular representation, evaluates a clock-shift determinant, and uses a cyclic recurrence to show that every singular nonlinear block has nullity one. Unit coefficients recover a characteristic-three jump for the $1 + a + b$ specialization. Full quotient matrices provide independent finite checks but are not used in the proof.

1. INTRODUCTION

Periodic configurations convert an infinite symbolic constraint into a finite linear problem. For shifts over an abelian lattice, Fourier characters then diagonalize convolution, and nullity becomes a count of torsion points on the zero set of the symbol. This positive-characteristic viewpoint is developed systematically by Zaidenberg [5]. The discrete Heisenberg group is a basic nonabelian nilpotent setting in which that character calculation is incomplete: most of the finite regular representation lies in nonlinear blocks.

For unit coefficients, the character resultant is Wendt's determinant, whose relation to finite-field points on Fermat curves is classical [1]. We therefore do not present the cyclotomic gcd itself as a new number-theoretic invariant. Its role is to isolate the abelian stratum before the nonlinear Heisenberg jump is restored.

We study the finite-field group shift defined by the weighted rule

$$(1) \quad \alpha x_g + \beta x_{ga} + \gamma x_{gb} = 0.$$

The underlying Heisenberg principal-action framework is established. In particular, Göll et al. [2] develop expansiveness and homoclinic methods for principal actions, and the survey of Lind and Schmidt [4] treats the exact integer element $1 + a + b$ as a mixing example. Our object is its zero-dimensional coefficient reduction, not a claim that the group, action framework, or three-term element is new. The question here is narrower and arithmetic: what is the exact kernel dimension after imposing a prime congruence period?

The answer has two qualitatively different pieces. The one-dimensional representations behave as in abelian harmonic analysis and contribute a polynomial gcd degree. The degree- ℓ representations see the clock-shift matrix

$$\alpha I + \beta U + \gamma V.$$

Its determinant is independent of the nontrivial central character and equals $\alpha^\ell + \beta^\ell + \gamma^\ell$. Thus all nonlinear types become singular together. Each singular block has nullity one, but its regular multiplicity is ℓ ; summing over the $\ell - 1$ nontrivial central characters creates a jump of size $\ell(\ell - 1)$.

The paper makes three concrete contributions.

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TABLE 1. The two representation strata in the fixed-space formula. A block nullity is multiplied by its multiplicity in the regular representation.

Stratum	Types	Degree	Multiplicity	Singular contribution
Characters	ℓ^2	1	1	$D_{p,\ell}(\alpha, \beta, \gamma)$
Nontrivial central character	$\ell - 1$	ℓ	ℓ	$\ell(\ell - 1)\mathbf{1}_{\{\Delta_\ell=0\}}$

- (1) We give an exact fixed-dimension formula for every nondegenerate weight triple and every pair of distinct primes p, ℓ , with ℓ odd.
- (2) We separate the character contribution from a nonlinear Fermat-locus jump and prove the latter with an exact corank-one statement, not only a determinant test.
- (3) We audit right versus left convolution and verify the formula against full ℓ^3 -dimensional matrices in ten independent cases.

Finite Heisenberg representation theory, including the Stone–von Neumann description behind the nonlinear blocks, is standard; we use the formulation of Gurevich and Hadani [3]. The progress here is the resulting symbolic fixed-space phase diagram for the weighted family. No priority language is intended: the literature search supporting this internal draft is bounded.

2. THE SHIFT, THE QUOTIENT, AND THE FORMULA

Write the discrete Heisenberg group in coordinates as $\Gamma = \mathbb{Z}^3$ with multiplication

$$(2) \quad (r, s, t)(u, v, w) = (r + u, s + v, t + w + rv).$$

Set $a = (1, 0, 0)$, $b = (0, 1, 0)$, and $c = (0, 0, 1)$, so that $[a, b] = c$ and c is central. For an odd prime ℓ , reduction of all three coordinates modulo ℓ gives

$$\pi_\ell : \Gamma \longrightarrow Q_\ell := \text{Heis}(\mathbb{F}_\ell), \quad N_\ell := \ker \pi_\ell.$$

Let $p \neq \ell$ be prime, and fix $\alpha, \beta, \gamma \in \mathbb{F}_p^\times$. On $\mathbb{F}_p^\Gamma$ use the left shift $(h \cdot x)_g = x_{h^{-1}g}$ and define

$$(3) \quad X_{p;\alpha,\beta,\gamma} = \{x : \alpha x_g + \beta x_{ga} + \gamma x_{gb} = 0 \text{ for every } g \in \Gamma\}.$$

This is a closed shift-invariant linear group shift.

An N_ℓ -fixed configuration is constant on left cosets. Normality of N_ℓ identifies those cosets with Q_ℓ , and (3) becomes the right-convolution kernel

$$(4) \quad (T_{\alpha,\beta,\gamma}f)(q) = \alpha f(q) + \beta f(qa) + \gamma f(qb), \quad f \in \mathbb{F}_p^{Q_\ell}.$$

Define

$$(5) \quad D_{p,\ell}(\alpha, \beta, \gamma) := \deg_{\mathbb{F}_p[t]} \gcd(t^\ell - 1, (\alpha + \beta t)^\ell + \gamma^\ell)$$

and

$$(6) \quad \Delta_\ell(\alpha, \beta, \gamma) := \alpha^\ell + \beta^\ell + \gamma^\ell \in \mathbb{F}_p.$$

Theorem 2.1 (Exact congruence-fixed dimension). *Under the assumptions above,*

$$(7) \quad \dim_{\mathbb{F}_p} \text{Fix}_{N_\ell} X_{p;\alpha,\beta,\gamma} = D_{p,\ell}(\alpha, \beta, \gamma) + \ell(\ell - 1)\mathbf{1}_{\{\Delta_\ell(\alpha,\beta,\gamma)=0\}}.$$

Equivalently, the number of N_ℓ -fixed configurations is p raised to the right-hand side of (7).

The two terms in (7) are summarized in table 1. Their different scales are essential: the character term is at most ℓ , whereas the nonlinear term is either zero or $\ell(\ell - 1)$.

3. FINITE HEISENBERG REGULAR DECOMPOSITION

Let k be an algebraic closure of $\mathbb{F}_p$. We first remove a possible field splitting ambiguity.

Lemma 3.1 (Base change). *For a matrix A over $\mathbb{F}_p$, its nullity over $\mathbb{F}_p$ equals the nullity of $A \otimes_{\mathbb{F}_p} k$ over k .*

Proof. Field extension is flat, so tensoring the exact sequence $0 \rightarrow \ker A \rightarrow \mathbb{F}_p^n \rightarrow \mathbb{F}_p^m$ with k remains exact. Equivalently, the nonzero minors determining the rank remain nonzero after embedding $\mathbb{F}_p \hookrightarrow k$. $\square$

The complex Stone–von Neumann theorem supplies the standard owner and model for finite Heisenberg representations [3]. Since our splitting field has characteristic p , we record the cross-characteristic classification directly rather than silently transferring the complex statement.

Lemma 3.2 (Cross-characteristic irreducibles). *Over k , the irreducibles of Q_ℓ are the following:*

- (1) *for every $(u, v) \in \mu_\ell(k)^2$, the character*

$$\chi_{u,v}(a) = u, \quad \chi_{u,v}(b) = v, \quad \chi_{u,v}(c) = 1;$$

- (2) *for every nontrivial $\zeta \in \mu_\ell(k)$, one degree- ℓ module π_ζ on the basis $e_0, \dots, e_{\ell-1}$, given by*

$$\pi_\zeta(a)e_j = \zeta^j e_j, \quad \pi_\zeta(b)e_j = e_{j+1 \bmod \ell}, \quad \pi_\zeta(c) = \zeta I.$$

The modules in this list are pairwise inequivalent and the list is complete.

Proof. Because $p \neq \ell$, the polynomial $t^\ell - 1$ has ℓ distinct roots in k . The characters in (1) are therefore distinct and exhaust the characters of the abelianization $Q_\ell/[Q_\ell, Q_\ell] \cong \mathbb{F}_\ell^2$.

For (2), the displayed clock and shift matrices satisfy $\pi_\zeta(a)\pi_\zeta(b) = \zeta\pi_\zeta(b)\pi_\zeta(a)$, matching $[a, b] = c$. The clock matrix has ℓ distinct one-dimensional eigenspaces. Any invariant subspace is a sum of some of them, because their spectral projectors are polynomials in the clock matrix. The shift matrix cyclically permutes all eigenspaces, so a nonzero invariant subspace contains all of them. Thus π_ζ is irreducible. Distinct values of ζ give inequivalent modules because the center acts by different scalars.

Maschke's theorem applies over k , and the sum of the squared degrees of the displayed pairwise inequivalent irreducibles is

$$\ell^2 + (\ell - 1)\ell^2 = \ell^3 = |Q_\ell|.$$

For a split semisimple group algebra this is the full dimension ledger, so there are no further irreducibles. $\square$

Proposition 3.3 (Regular-module ledger). *Over k , the right regular module of Q_ℓ decomposes into the ℓ^2 characters, each once, and the $\ell - 1$ modules π_ζ , each with multiplicity ℓ . Consequently,*

$$(8) \quad \begin{aligned} \text{null}(T_{\alpha, \beta, \gamma}) &= \#\{(u, v) \in \mu_\ell(k)^2 : \alpha + \beta u + \gamma v = 0\} \\ &+ \ell \sum_{\substack{\zeta^\ell = 1 \\ \zeta \neq 1}} \text{null}(\alpha I + \beta \pi_\zeta(a) + \gamma \pi_\zeta(b)). \end{aligned}$$

Proof. Maschke's theorem applies because $p \nmid |Q_\ell|$, and lemma 3.2 gives the complete split list. To freeze the right-translation convention, set $(R_h f)(q) = f(qh)$. For a representation π on V , identify its matrix-coefficient space with $V^* \otimes V$ by

$$\varphi_{\lambda, v}(q) = \lambda(\pi(q)v).$$

Then

$$R_h \varphi_{\lambda, v}(q) = \lambda(\pi(q)\pi(h)v) = \varphi_{\lambda, \pi(h)v}(q).$$

Thus R_h acts as $I_{V^*} \otimes \pi(h)$: the right regular module contains $\dim V$ copies of V , and the finite operator acts on each copy by $\alpha I + \beta \pi(a) + \gamma \pi(b)$. Inserting the degrees from lemma 3.2 proves (8). $\square$

Remark 3.4 (Right versus left). Our finite operator is right translation because the local rule reads q, qa, qb , and the matrix-coefficient calculation above gives the displayed $\pi(a), \pi(b)$ blocks without a transpose or inverse. Under the common dual convention the blocks are contragredient. On characters this replaces (u, v) by (u^{-1}, v^{-1}) , a permutation of $\mu_\ell(k)^2$. On nonlinear modules it replaces the central scalar ζ by ζ^{-1} , a permutation of the nontrivial central characters. Hence the summed nullities in (8) are unchanged even though individual matrices need not be identical.

4. THE CHARACTER CONTRIBUTION

The first line of (8) is a commutative torsion-point count. The following calculation identifies it over the ground field and also fixes the sign in (5).

Proposition 4.1 (Cyclotomic intersection). *The number of singular character blocks is $D_{p,\ell}(\alpha, \beta, \gamma)$.*

Proof. For $u, v \in \mu_\ell(k)$, singularity means

$$(9) \quad \alpha + \beta u + \gamma v = 0.$$

Because $\gamma \neq 0$, a choice of u determines the only possible value

$$v = -\frac{\alpha + \beta u}{\gamma}.$$

Since ℓ is odd, the condition $v^\ell = 1$ becomes

$$-\frac{(\alpha + \beta u)^\ell}{\gamma^\ell} = 1, \quad \text{or equivalently} \quad (\alpha + \beta u)^\ell + \gamma^\ell = 0.$$

Thus singular characters are in bijection with the common roots of the two polynomials in (5). The polynomial $t^\ell - 1$ is separable in characteristic p because $p \neq \ell$. Hence the degree of the gcd counts those common roots once each. $\square$

Remark 4.2 (Arithmetic content). The term $D_{p,\ell}$ is an intersection of the ℓ -torsion torus with the affine curve $\alpha + \beta u + \gamma v = 0$. It satisfies $0 \leq D_{p,\ell} \leq \ell$ and can vary with both primes and the coefficient point. This is the part of the answer analogous to lattice Fourier analysis in positive characteristic [5]. The next section accounts for the nonabelian mass absent from that calculation. For $(\alpha, \beta, \gamma) = (1, 1, 1)$, the corresponding resultant is the classical Wendt determinant [1]; we use the gcd only as a block count and claim no new resultant theory.

5. CLOCK-SHIFT BLOCKS AND THE FERMAT JUMP

Fix a nontrivial central character. Use the explicit module π_ζ from lemma 3.2; writing its two generators again,

$$(10) \quad U = \text{diag}(1, \zeta, \dots, \zeta^{\ell-1}), \quad V e_j = e_{j+1 \bmod \ell}.$$

Changing the central character only replaces ζ by another primitive root, which will not change the calculation below.

Lemma 5.1 (Clock-shift determinant). *For nonzero $\alpha, \beta, \gamma \in k$,*

$$(11) \quad \det(\alpha I + \beta U + \gamma V) = \alpha^\ell + \beta^\ell + \gamma^\ell.$$

Proof. Put $d_j = \alpha + \beta \zeta^j$. In the determinant of $\text{diag}(d_0, \dots, d_{\ell-1}) + \gamma V$, a nonzero permutation term can use only the full diagonal or the unique full ℓ -cycle. The cycle sign is $(-1)^{\ell-1} = 1$ because ℓ is odd. Therefore

$$\det(\alpha I + \beta U + \gamma V) = \prod_{j=0}^{\ell-1} (\alpha + \beta \zeta^j) + \gamma^\ell.$$

The factorization of $X^\ell - Y^\ell$, evaluated at $X = \alpha$ and $Y = -\beta$, gives $\prod_j (\alpha + \beta \zeta^j) = \alpha^\ell + \beta^\ell$. This proves (11). $\square$

A determinant alone would not determine the fixed-space dimension. The special cyclic sparsity gives the required corank statement.

Lemma 5.2 (Exact nonlinear nullity). *Every nonlinear block is invertible when $\Delta_\ell \neq 0$ and has nullity exactly one when $\Delta_\ell = 0$.*

Proof. Since $(Vx)_j = x_{j-1}$, in the basis of (10) the equation $(\alpha I + \beta U + \gamma V)x = 0$ is exactly (12)

$$(\alpha + \beta \zeta^j)x_j + \gamma x_{j-1} = 0 \quad (j \bmod \ell).$$

Since $\gamma \neq 0$, a single coordinate determines all other coordinates by successive use of (12). The kernel therefore has dimension at most one. By lemma 5.1, it is nonzero exactly when $\Delta_\ell = 0$, in which case its dimension is one. $\square$

Proof of theorem 2.1. By proposition 4.1, the characters contribute $D_{p,\ell}$. By lemma 5.2, either none or all of the $\ell - 1$ nonlinear types are singular. Each singular type has block nullity one and multiplicity ℓ in the regular representation, so its total contribution is $\ell(\ell - 1)$. Insert these terms into (8), then use lemma 3.1 to descend the nullity from k to $\mathbb{F}_p$. $\square$

6. COEFFICIENT PHASE DIAGRAM AND FINITE CONTROLS

Multiplying (α, β, γ) by a nonzero scalar multiplies the finite operator by that scalar and does not change its kernel. The family therefore lives naturally in the projective coefficient plane.

Corollary 6.1 (Projective phase diagram). *On $\mathbb{P}^2(\mathbb{F}_p)$ away from the coordinate axes, the nonlinear part of the fixed dimension is zero off the Fermat curve*

$$\alpha^\ell + \beta^\ell + \gamma^\ell = 0$$

and equals $\ell(\ell - 1)$ on that curve. The remaining variation is the bounded term $D_{p,\ell} \leq \ell$.

Proof. The Fermat equation is homogeneous, and scaling both polynomials in (5) by a nonzero ℓ th power does not change their common roots. The claim now follows from theorem 2.1. $\square$

Corollary 6.2 (The unit-coefficient shift). *For the rule $x_g + x_{ga} + x_{gb} = 0$,*

$$\dim_{\mathbb{F}_p} \text{Fix}_{N_\ell} X_{p;1,1,1} = \deg_{\mathbb{F}_p[t]} \gcd(t^\ell - 1, (-1 - t)^\ell - 1) + \ell(\ell - 1)\mathbf{1}_{\{p=3\}}.$$

Proof. For odd ℓ , $(-1 - t)^\ell - 1 = -((1 + t)^\ell + 1)$, so the two gcd expressions agree up to a unit. The nonlinear determinant is $1 + 1 + 1 = 3$, which vanishes in a prime field exactly in characteristic three. $\square$

We checked the formula by constructing the full matrix of (4) in the coordinate model (2). Gaussian elimination was performed over the indicated prime field. A selection of the frozen results appears in table 2; the complete receipt and code accompany the paper. The control also constructs four clock-shift blocks directly over fields containing the required roots of unity: two singular and two nonsingular cases verify both lemma 5.1 and the zero/one nullity dichotomy in lemma 5.2.

The full quotient matrices verify the implementation of the displayed group law, the selected finite operator, and the final nullity formula on the sample tuples. They can expose many transcription or implementation mistakes, including an omitted regular multiplicity. Nullity comparison alone does not distinguish the selected right-translation convention from the dual left convention, because remark 3.4 shows that the total nullity is invariant under that change. All computations remain finite regression evidence and are not used to infer the all-prime theorem.

TABLE 2. Independent full-matrix controls. In every row the observed nullity equals (7).

ℓ	p	(α, β, γ)	Fermat stratum	Nullity
3	2	(1, 1, 1)	nonsingular	2
3	5	(1, 1, 2)	singular	6
3	5	(1, 2, 3)	nonsingular	0
3	7	(2, 3, 4)	nonsingular	1
5	3	(1, 1, 1)	singular	21
5	11	(1, 1, 1)	nonsingular	3
5	11	(2, 3, 5)	nonsingular	2

7. SCOPE, LIMITATIONS, AND DECLARATIONS

Ownership and scope. Principal algebraic actions of the discrete Heisenberg group and the exact integer element $1 + a + b$ precede this paper [2, 4]. Finite Heisenberg Stone–von Neumann theory is also standard [3]. We claim only the weighted mod- p congruence-nullity formula proved here. A bounded search found no exact match, but that is not a worldwide novelty or priority certificate.

Limitations. The assumption $p \neq \ell$ is structural: at $p = \ell$, the finite regular module need not be semisimple, and nilpotent extensions can change the nullity. We also require all three coefficients to be nonzero. Degenerate two-term rules admit separate elementary case splits but are outside the selected theorem. Finally, congruence-fixed dimensions do not by themselves settle expansiveness, entropy, density of periodic points, or generator questions for the infinite action.

Data and code availability. No external data are used. The finite-control source and its frozen text receipt are included in the manuscript directory. They generate every reported matrix nullity from the displayed group law and coefficient tuple.

Ethics declaration. The work is purely mathematical, uses no human participants, animals, or sensitive data, and requires no ethics approval.

Author contributions. Conceptualization, formal analysis, software, verification, writing—original draft, and writing—review and editing were performed by the anonymous author or authors. Attribution must be replaced by the final CRediT record before external submission. Conflicts of interest and funding. No conflict of interest or external funding is declared in this internal draft.

Tool-use disclosure. Automated tools assisted with finite regression checks, source discovery, LaTeX compilation, and language editing. The formal arguments and all bibliographic metadata require author verification before external use.

8. CONCLUSION

Weighted three-term Heisenberg shifts have an exact finite-quotient phase diagram. Character blocks contribute the cyclotomic intersection degree $D_{p,\ell}$, while every nonlinear block is controlled by the same Fermat polynomial $\alpha^\ell + \beta^\ell + \gamma^\ell$. The regular multiplicities turn its vanishing into a jump of $\ell(\ell - 1)$. The unit-coefficient characteristic-three phenomenon is therefore one specialization of a coefficient-space theorem rather than an isolated determinant coincidence.

The excluded modular case $p = \ell$ is the next concrete problem. There the failure of semisimplicity should replace the two-stratum ledger by a radical filtration, so the present formula identifies both the baseline and the exact point at which new representation-theoretic behavior must enter.

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